

Splunk Enterprise SIEM Deployment & Configuration Guide

Enterprise Network Security Monitoring Implementation

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Scope: Splunk SIEM deployment with network device integration (pfSense firewall + Cisco switches)

Table of Contents

1. [Executive Summary](#)
 2. [Architecture Overview](#)
 3. [Prerequisites](#)
 4. [Part 1: Splunk Server Deployment](#)
 5. [Part 2: Splunk Configuration](#)
 6. [Part 3: pfSense Firewall Integration](#)
 7. [Part 4: Cisco Switch Integration](#)
 8. [Part 5: Verification & Testing](#)
 9. [Security Best Practices](#)
 10. [Troubleshooting](#)
 11. [Appendix](#)
-

Executive Summary

This document provides a complete, step-by-step guide for deploying Splunk Enterprise as a Security Information and Event Management (SIEM) solution for enterprise network security monitoring. The implementation follows Zero Trust security principles and integrates centralized logging from network infrastructure devices.

What This Guide Covers

- **Splunk Enterprise deployment** on Ubuntu 24.04 LTS
- **Index creation** for organized log storage
- **Data input configuration** for syslog reception
- **pfSense firewall** log forwarding
- **Cisco switch** secure logging with SSH hardening
- **Zero Trust access controls** via Jump Server architecture

Implementation Results

Infrastructure Deployed:

- 1 Splunk Enterprise SIEM server (10.0.0.75)
- 1 pfSense HA firewall cluster (10.0.0.1/2)
- 4 Cisco switches with centralized logging

Logs Collected (First 24 Hours):

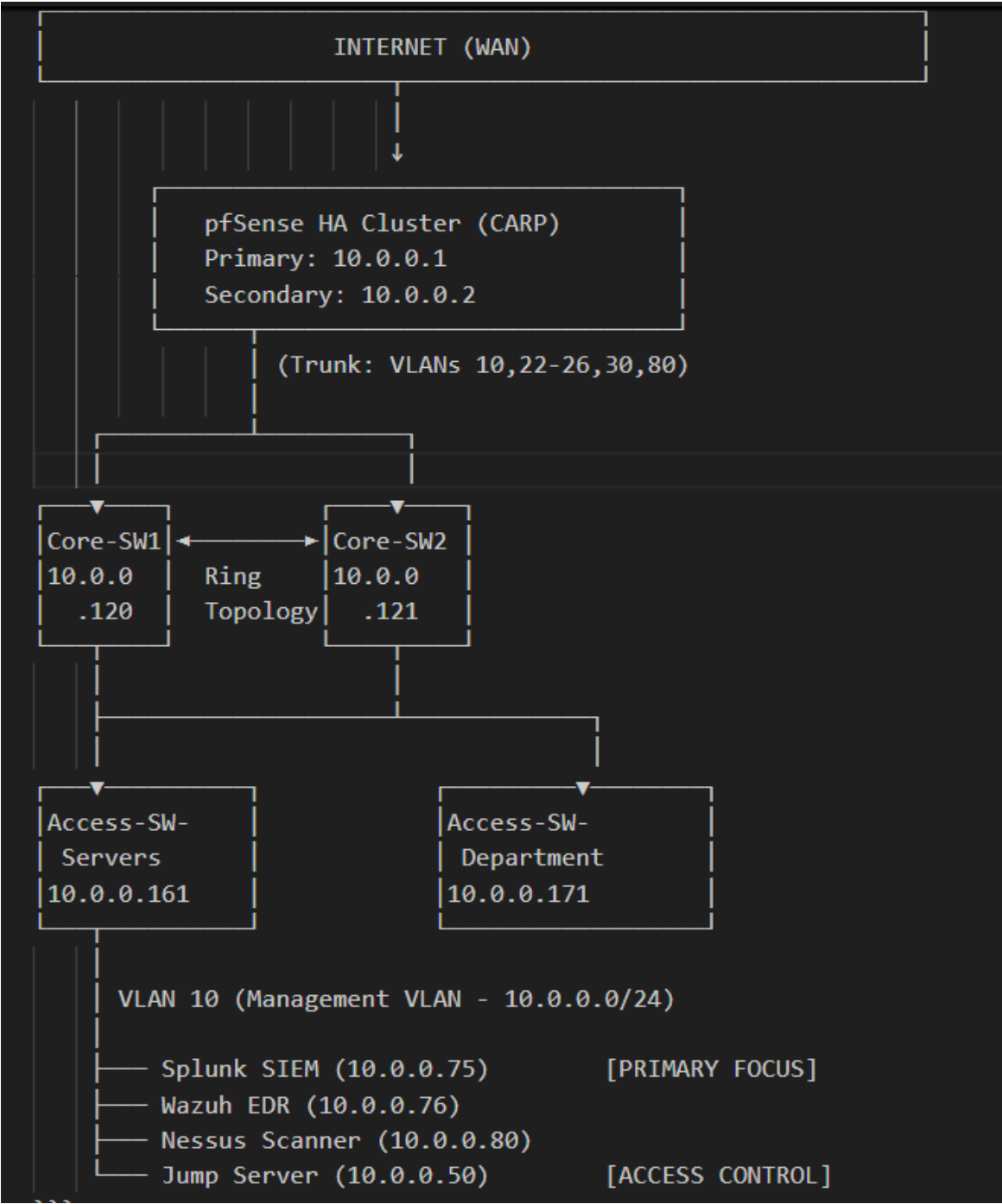
- 30,977 firewall events from pfSense
- System and configuration events from 4 network switches
- All logs indexed and searchable in real-time

Security Posture:

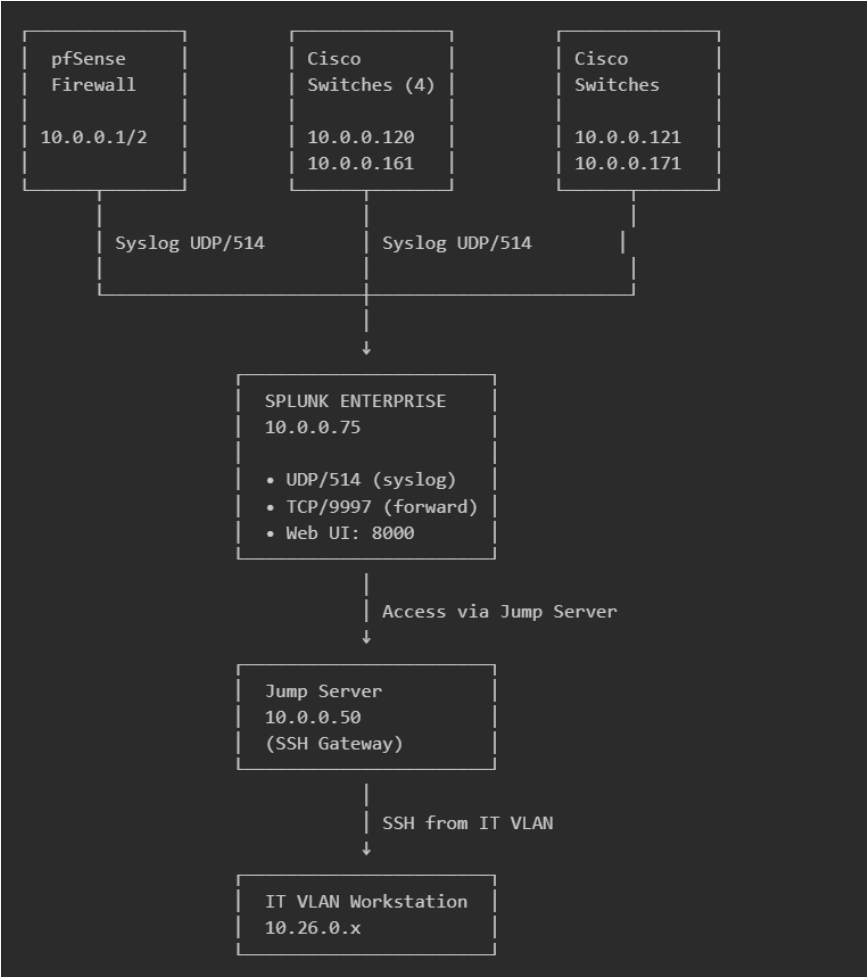
- Zero Trust architecture: All management via Jump Server
 - SSH hardened with ACLs restricting access
 - Encrypted log transmission
 - Comprehensive audit trail
-

Architecture Overview

Network Topology



Data Flow Architecture



VLAN Architecture

VLAN ID	Name	Subnet	Purpose
10	Management	10.0.0.0/24	Security monitoring infrastructure
22	Sales	10.22.0.0/25	Sales department workstations
23	Finance	10.23.0.0/25	Finance department workstations
24	HR	10.24.0.0/25	HR department workstations
26	IT	10.26.0.0/25	IT department workstations
30	Servers	10.30.0.0/24	Domain controllers, file servers
80	Storage	10.80.0.0/24	Storage network

Prerequisites

Hardware Requirements (Splunk Server)

Component	Minimum	Recommended	Implemented
CPU	4 cores	6 cores	6 cores
RAM	8 GB	16 GB	16 GB

Component	Minimum	Recommended	Implemented
Disk	100 GB	250 GB	250 GB
Network	1 Gbps NIC	1 Gbps NIC	1 Gbps

Note: Disk space scales with log volume. Plan for ~500 MB/day for this network size.

Software Requirements

Software	Version	Purpose
Ubuntu Server	24.04 LTS	Operating system
Splunk Enterprise	10.0.2+	SIEM platform
pfSense	2.7+	Firewall (log source)
Cisco IOS	15.2+	Switches (log sources)

Network Requirements

- **IP Address:** Static IP in Management VLAN (10.0.0.75/24)
- **Gateway:** pfSense firewall (10.0.0.1)
- **DNS:** Domain Controller (10.30.0.10) or public DNS (8.8.8.8)
- **Firewall Rules:**
 - UDP/514 (syslog receiving)
 - TCP/8000 (web interface access from IT VLAN)
 - TCP/9997 (Splunk forwarder receiving)

Access Requirements

- **Jump Server access:** SSH to 10.0.0.50
- **Administrative credentials:** For Splunk installation
- **pfSense admin access:** For log configuration
- **Switch console/SSH access:** For configuration

Part 1: Splunk Server Deployment

Step 1.1: Deploy Ubuntu Server VM

1. Create VM in virtualization platform (EVE-NG/VMware/KVM):

- **Name:** Splunk-SIEM

- **OS:** Ubuntu Server 24.04 LTS
- **vCPU:** 6 cores
- **RAM:** 16 GB
- **Disk:** 250 GB
- **Network:** Disconnected (initially)

2. Install Ubuntu Server 24.04:

- Boot from ISO
- Select language: English
- Keyboard layout: US
- Network: Skip (configure later)
- Storage: Use entire disk
- Profile setup:
 - **Name:** Splunk Administrator
 - **Server name:** splunk
 - **Username:** ubuntu (or your preference)
 - **Password:** Strong password (store securely)
- SSH server: Install OpenSSH server ✓
- Featured snaps: None
- Complete installation and reboot

3. Update the system:

```
bash  
  
sudo apt update  
sudo apt upgrade -y
```

Step 1.2: Download Splunk Enterprise

Option A: Download on VM (requires temporary internet connection)

```
bash
```

```
cd /tmp
```

```
# Download Splunk Enterprise (version may vary)
```

```
wget -O splunk-10.0.2-e2d18b476e9-Linux-amd64.tgz \
```

```
'https://download.splunk.com/products/splunk/releases/10.0.2/linux/splunk-10.0.2-e2d18b476e9-Linux-amd64.tgz'
```

```
# Verify download size (~60-70 MB)
```

```
ls -lh splunk-10.0.2-e2d18b476e9-Linux-amd64.tgz
```

Option B: Download on host machine and transfer via SCP

1. Download from https://www.splunk.com/en_us/download/splunk-enterprise.html
2. Select: Linux → 64-bit → .tgz
3. Transfer to VM:

```
bash
```

```
scp splunk-10.0.2-*.tgz ubuntu@<vm-temp-ip>:/tmp/
```

Step 1.3: Install Splunk Enterprise

```
bash
```

```
# Navigate to /opt
```

```
cd /opt
```

```
# Extract Splunk tarball
```

```
sudo tar xvf /tmp/splunk-10.0.2-e2d18b476e9-Linux-amd64.tgz
```

```
# Verify extraction
```

```
ls -ld /opt/splunk
```

```
# Should show: drwxr-xr-x splunk splunk
```

Step 1.4: Start Splunk and Accept License

```
bash
```

```
# Start Splunk for the first time
```

```
cd /opt/splunk/bin
```

```
sudo ./splunk start --accept-license
```

You will be prompted to create admin credentials:

Please enter an administrator username: admin
Please enter a new password: [Enter strong password]
Please confirm new password: [Re-enter password]

Important: Save these credentials securely. You'll need them for web UI access.

Splunk will start and display:

Waiting for web server at http://127.0.0.1:8000 to be available.... Done

The Splunk web interface is at http://splunk:8000

Step 1.5: Enable Splunk to Start on Boot

```
bash  
  
sudo /opt/splunk/bin/splunk enable boot-start
```

Output:

Init script installed at /etc/init.d/splunk.

Step 1.6: Configure Splunk to Listen on All Interfaces

By default, Splunk web interface listens on `127.0.0.1` (localhost only). We need it accessible from the network.

Check current configuration:

```
bash  
  
sudo ss -tlnp | grep 8000
```

If it shows `127.0.0.1:8000`, reconfigure:

```
bash  
  
# Set web interface to listen on all interfaces  
sudo /opt/splunk/bin/splunk set web-port 8000 -httpport 8000:0.0.0.0  
  
# Enter admin credentials when prompted  
  
# Restart Splunk web  
sudo /opt/splunk/bin/splunk restart splunkweb
```

Verify:


```
bash
```

```
sudo ss -tlnp | grep 8000
```

Should show: `LISTEN 0 128 0.0.0.0:8000 0.0.0.0:*`

Step 1.7: Configure Static IP Address

1. Check network interface name:

```
bash
```

```
ip a
```

Look for interface (usually `ens3`, `eth0`, or similar)

2. Edit netplan configuration:

```
bash
```

```
sudo nano /etc/netplan/00-installer-config.yaml
```

3. Configure static IP:

```
yaml
```

```
network:
  version: 2
  ethernets:
    ens3: # Replace with your interface name
      dhcp4: no
      addresses:
        - 10.0.0.75/24
      routes:
        - to: default
          via: 10.0.0.1
      nameservers:
        addresses:
          - 10.30.0.10 # DC01 (internal DNS)
          - 8.8.8.8 # Google DNS (backup)
```

4. Test configuration:

```
bash
```

```
sudo netplan try
```

Press **Enter** to accept if configuration is valid.

5. Apply permanently:

```
bash  
  
sudo netplan apply
```

6. Verify IP address:

```
bash  
  
ip a show ens3
```

Should show: `inet 10.0.0.75/24`

Step 1.8: Connect to Management VLAN

In virtualization environment (EVE-NG example):

1. Stop the VM (or keep running)
2. Connect VM network interface to **Access-SW-Servers** (or appropriate switch)
3. Configure switch port for **VLAN 10** (Management)

Switch configuration (Cisco IOS):

```
cisco  
  
enable  
configure terminal  
  
interface ethernet 2/1  
description Splunk-SIEM-Server  
switchport mode access  
switchport access vlan 10  
spanning-tree portfast  
no shutdown  
exit  
  
write memory
```

4. Test connectivity:

```
bash
```

```
# Test gateway
```

```
ping -c 3 10.0.0.1
```

```
# Test DNS
```

```
ping -c 3 10.30.0.10
```

```
# Test internet (if routed)
```

```
ping -c 3 8.8.8.8
```

Step 1.9: Configure Firewall Rules (pfSense)

Allow IT VLAN to access Splunk web interface.

In pfSense Web GUI:

1. Navigate to **Firewall** → **Rules** → **VLAN26_IT_VLAN**
2. Click + **Add** (add to top)
3. Configure rule:
 - **Action:** Pass
 - **Interface:** VLAN26_IT_VLAN
 - **Protocol:** TCP
 - **Source:** VLAN26_IT_VLAN subnets
 - **Destination:** Single host → 10.0.0.75
 - **Destination Port:** HTTP (Other) → Custom: 8000
 - **Description:** Allow IT VLAN to access Splunk Web Interface (port 8000)
4. Click **Save**
5. Click **Apply Changes**

Step 1.10: Verify Web Interface Access

From IT VLAN workstation (10.26.0.x):

1. Open web browser
2. Navigate to: http://10.0.0.75:8000
3. Login with admin credentials created in Step 1.4

You should see the Splunk Enterprise home page.

Part 2: Splunk Configuration

Step 2.1: Create Indexes for Log Organization

Indexes separate different log types for efficient searching and retention management.

1. Access Splunk Web UI: `http://10.0.0.75:8000`

2. Navigate to Settings → Indexes

3. Click "New Index" (top right)

4. Create the following indexes:

Index 1: network

- Index Name: `network`
- Index Data Type: Events
- Max Size of Index (MB): `5000` (5 GB)
- Frozen Path: (leave default)
- Searchable Days: 90
- Click Save

Index 2: windows

- Index Name: `windows`
- Max Size of Index (MB): `5000` (5 GB)
- Click Save

Index 3: linux

- Index Name: `linux`
- Max Size of Index (MB): `5000` (5 GB)
- Click Save

Index 4: wazuh

- Index Name: `wazuh`
- Max Size of Index (MB): `5000` (5 GB)
- Click Save

5. Verify indexes:

Navigate back to **Settings** → **Indexes**

You should see all 4 indexes listed with status "Enabled".

Index Summary:

Index Name	Purpose	Retention	Size Limit
network	pfSense, switches, network devices	90 days	5 GB
windows	Windows servers, domain controllers	90 days	5 GB
linux	Linux servers, Jump Server	90 days	5 GB
wazuh	Wazuh EDR alerts and events	90 days	5 GB

Step 2.2: Configure Data Inputs - UDP Syslog (Port 514)

Configure Splunk to receive syslog messages from network devices.

1. Navigate to **Settings** → **Data inputs**
2. Find **"UDP"** in the **"Local inputs"** section
3. Click **" + Add new "** next to **UDP**
4. Configure **UDP input**:

Page 1 - Select Source:

- **Port:**
- Click **Next**

Page 2 - Input Settings:

- **Source type:** Select **"syslog"** from dropdown
- **App Context:** Search & Reporting
- **Index:** Select **"network"**
- Leave other fields as default
- Click **Review**

Page 3 - Review:

- Verify settings:
 - Input Type: UDP Port
 - Port Number: 514
 - Source Type: syslog

- Index: network
- Click **Submit**

5. Confirmation:

You should see: "UDP input has been created successfully"

6. Verify UDP listener:

SSH to Splunk server and check:

```
bash  
  
sudo ss -ulnp | grep 514
```

Should show: `LISTEN 0.0.0.0:514` (Splunk listening on all interfaces)

Step 2.3: Configure Data Inputs - TCP Forwarders (Port 9997)

Configure Splunk to receive data from Splunk Universal Forwarders.

1. Navigate to Settings → Forwarding and receiving

2. Click "Configure receiving"

3. Click "New Receiving Port"

4. Configure TCP input:

- **Listen on this port:** `9997`
- Click **Save**

5. Verification:

You should see port 9997 listed with status "Enabled"

6. Verify TCP listener:

```
bash  
  
sudo ss -tlnp | grep 9997
```

Should show: `LISTEN 0.0.0.0:9997`

Step 2.4: Verify Splunk Configuration

Check all configured inputs:

```
bash
```

```
# From Splunk server
sudo /opt/splunk/bin/splunk list inputstatus

# Or via Web UI:
# Settings → Data inputs
```

Expected configuration:

- ☒ UDP port 514 listening (syslog → network index)
- ☒ TCP port 9997 listening (forwarders)
- ☒ 4 indexes created (network, windows, linux, wazuh)
- ☒ Web interface on port 8000
- ☒ Management interface on port 8089

Splunk server is now ready to receive logs!

Part 3: pfSense Firewall Integration

Step 3.1: Access pfSense Web Interface

Via Jump Server (recommended for production):

```
bash

# From your workstation, SSH to Jump Server
ssh ubuntu@10.0.0.50

# From Jump Server, SSH to pfSense
ssh admin@10.0.0.1
```

Or via web GUI (if accessible):

```
https://10.0.0.1
```

Step 3.2: Configure Remote Logging

1. Navigate to Status → System Logs → Settings
2. Scroll to "Remote Logging Options" section
3. Configure syslog forwarding:

Enable Remote Logging:

- ☒ Check "Send log messages to remote syslog server"

Remote log servers:

- Enter:
- (Format: IP:PORT)

Source Address:

- Select: "**Default (any)**"
- (pfSense will use appropriate interface)

IP Protocol:

- Select: "**IPv4**"

Remote Syslog Contents - Select the following:

- ☒ **System Events** (system messages, configuration changes)
- ☒ **Firewall Events** (blocked/allowed traffic, filter log)
- ☒ **DHCP Events** (DHCP leases, releases)
- ☐ DNS Events (optional - can generate high volume)
- ☐ VPN Events (optional - if VPN is used)

4. Scroll down and click "Save"

Step 3.3: Verify pfSense Logging

1. Check pfSense system logs:

Navigate to **Status** → **System Logs** → **System**

Look for message:

2. Generate test traffic:

- Browse to any website from a workstation
- Block some traffic (firewall should log blocks)
- Request DHCP lease (renew IP on a workstation)

3. Verify in Splunk:


```
index=network source="udp:514" host="10.0.0.1"
| head 10
```

Expected results:

- Firewall filter logs (blocked/allowed traffic)
- System events (configuration changes)
- DHCP events (if enabled)

Sample pfSense log entry:

```
Dec 5 21:26:35 10.0.0.1 filterlog[63450]:
4,,1000000103,vnet0.30,match,block,in,4,0x0,,128,46572,0,none,17,udp,73,10.30.0.10,1.1.1.1,64,53,53
```

Fields:

- `10.0.0.1` = pfSense IP
 - `filterlog` = Firewall log type
 - `block` = Action taken
 - `10.30.0.10` = Source IP
 - `1.1.1.1` = Destination IP
-

Part 4: Cisco Switch Integration

Overview

This section covers:

1. SSH hardening for secure remote access
2. Access Control Lists (ACLs) for Jump Server-only access
3. Syslog configuration for centralized logging
4. Implementation across 4 switches

Switches to configure:

- Access-SW-Servers (10.0.0.161)
- Core-SW1 (10.0.0.120)
- Core-SW2 (10.0.0.121)

- Access-SW-Department (10.0.0.171)

Step 4.1: Prerequisites - Jump Server SSH Client Configuration

Cisco IOS devices use older SSH algorithms that may not be enabled by default on modern SSH clients.

On Jump Server (10.0.0.50):

```
bash

# Edit SSH client configuration
sudo nano /etc/ssh/ssh_config
```

Add at the end of the file:

```
# Cisco IOS compatibility
Host 10.0.0.*
    KexAlgorithms +diffie-hellman-group14-sha1,diffie-hellman-group1-sha1
    HostKeyAlgorithms +ssh-rsa
    Ciphers +aes128-ctr,aes192-ctr,aes256-ctr
```

Save and exit (Ctrl+X, Y, Enter)

This enables legacy SSH algorithms required for Cisco IOS devices.

Step 4.2: Configure SSH + Syslog on Cisco Switches

Repeat the following steps for each switch.

Switch 1: Access-SW-Servers (10.0.0.161)

1. Console into the switch (via EVE-NG or physical console)

2. Enter configuration mode:

```
cisco

enable
configure terminal
```

3. Configure SSH prerequisites:

```
cisco
```

```
! Set domain name (required for SSH key generation)
```

```
ip domain-name company.local
```

```
! Generate RSA cryptographic keys (2048-bit for security)
```

```
crypto key generate rsa modulus 2048
```

```
! Output: "Generating 2048 bit RSA keys, keys will be non-exportable...[OK]"
```

```
! This takes ~30 seconds
```

4. Create administrative user:

```
cisco
```

```
! Create admin user with privilege level 15 (full access)
```

```
username admin privilege 15 secret SwitchAdmin2025
```

Note: Replace `SwitchAdmin2025` with your strong password.

5. Configure SSH hardening:

```
cisco
```

```
! Force SSH version 2 only (v1 is insecure)
```

```
ip ssh version 2
```

```
! Set SSH timeout to 60 seconds
```

```
ip ssh time-out 60
```

```
! Limit authentication retries to 2
```

```
ip ssh authentication-retries 2
```

6. Create Access Control List (ACL) - Jump Server Only:

```
cisco
```

```
! Create ACL to allow only Jump Server SSH access
```

```
access-list 10 remark ALLOW-JUMP-SERVER-ONLY
```

```
access-list 10 permit host 10.0.0.50
```

```
access-list 10 deny any log
```

Explanation:

- Line 1: Documentation comment
- Line 2: **Permit** SSH from Jump Server (10.0.0.50)

- Line 3: **Deny** all other IPs and **log** the attempts

7. Secure VTY lines (SSH access):

```
cisco

line vty 0 4
transport input ssh      ! Only allow SSH (block Telnet)
access-class 10 in       ! Apply ACL 10 to incoming connections
login local              ! Use local username/password database
exec-timeout 10 0       ! Auto-logout after 10 minutes of inactivity
logging synchronous     ! Prevent log messages from interrupting commands
exit
```

8. Configure syslog to Splunk:

```
cisco

! Set syslog destination
logging host 10.0.0.75

! Set logging level (informational = level 6)
! Levels: 0=emergencies, 1=alerts, 2=critical, 3=errors, 4=warnings, 5=notifications, 6=informational, 7=debugging
logging trap informational

! Set source interface for syslog packets
logging source-interface Vlan10

! Enable logging
logging on
```

Expected output:

```
*Dec 6 09:20:10.820: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.0.0.75 port 514 started - CLI initiated
```

9. Save configuration:

```
cisco

end

write memory
```

Expected output:

Building configuration...

[OK]

10. Verify configuration:

cisco

! Check logging configuration

show running-config | include logging

! Expected output:

! logging source-interface Vlan10

! logging host 10.0.0.75

! logging synchronous (under line vty)

! Check VTY line security

show running-config | section line vty

! Expected output:

! line vty 0 4

! access-class 10 in

! logging synchronous

! login local

! transport input ssh

! Check Access Control List

show access-lists 10

! Expected output:

! Standard IP access list 10

! 10 permit 10.0.0.50

! 20 deny any log

Switch 2: Core-SW1 (10.0.0.120)

Repeat Step 4.2 configuration commands (Steps 2-10) on Core-SW1.

Management IP: 10.0.0.120

Verify before proceeding to next switch.

Switch 3: Core-SW2 (10.0.0.121)

Repeat Step 4.2 configuration commands (Steps 2-10) on Core-SW2.

Management IP: 10.0.0.121

Switch 4: Access-SW-Department (10.0.0.171)

Repeat Step 4.2 configuration commands (Steps 2-10) on Access-SW-Department.

Management IP: 10.0.0.171

Optional: Set hostname first:

```
cisco  
  
configure terminal  
hostname Access-SW-Department
```

Step 4.3: Test SSH Access from Jump Server

From Jump Server (10.0.0.50):

```
bash  
  
# Test SSH to each switch  
ssh admin@10.0.0.161 # Access-SW-Servers  
ssh admin@10.0.0.120 # Core-SW1  
ssh admin@10.0.0.121 # Core-SW2  
ssh admin@10.0.0.171 # Access-SW-Department
```

First connection to each switch:

```
The authenticity of host '10.0.0.161 (10.0.0.161)' can't be established.  
RSA key fingerprint is SHA256:...  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Type and press Enter.

Enter password:

Expected result:

```
Access-SW-Servers>
```

Type to disconnect.

Step 4.4: Test SSH Access Restriction

Verify ACL blocks unauthorized access.

From IT VLAN workstation (NOT Jump Server):

```
bash
```

```
ssh admin@10.0.0.161
```

Expected result: Connection timeout or refused

```
ssh: connect to host 10.0.0.161 port 22: Connection timed out
```

This confirms ACL is working - only Jump Server can SSH to switches.

Check switch logs for denied attempt:

```
cisco
```

```
! On the switch console
```

```
show logging | include Deny
```

Should show:

```
%SEC-6-IPACCESSLOGP: list 10 denied tcp 10.26.0.100(54321) -> 10.0.0.161(22), 1 packet
```

Step 4.5: Verify Syslog in Splunk

1. In Splunk, run this search:

```
index=network source="udp:514"  
| stats count by host  
| sort - count
```

Expected results:

host	count
10.0.0.1	30,977 (pfSense)
10.0.0.161	6 (Access-SW-Servers)
10.0.0.120	3 (Core-SW1)
10.0.0.121	3 (Core-SW2)
10.0.0.171	3 (Access-SW-Department)

2. View sample switch log:

```
index=network host="10.0.0.161"  
| head 1
```

Sample switch log entry:

```
Dec 6 09:20:10.820: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.0.0.75 port 514 started - CLI initiated
```

3. Search for SSH access attempts:

```
index=network (host="10.0.0.161" OR host="10.0.0.120" OR host="10.0.0.121" OR host="10.0.0.171")
| search SSH OR "SEC-6-IPACCESSLOGP"
```

This shows all SSH-related events and ACL violations.

Part 5: Verification & Testing

Step 5.1: Verify All Data Sources

Run comprehensive verification search in Splunk:

```
index=network
| stats count by host, sourcetype
| sort host
```

Expected output:

host	sourcetype	count
10.0.0.1	syslog	30,977
10.0.0.120	syslog	3
10.0.0.121	syslog	3
10.0.0.161	syslog	6
10.0.0.171	syslog	3

Verification checklist:

- ✔ pfSense (10.0.0.1) - Firewall events
- ✔ Access-SW-Servers (10.0.0.161) - Switch logs
- ✔ Core-SW1 (10.0.0.120) - Switch logs
- ✔ Core-SW2 (10.0.0.121) - Switch logs
- ✔ Access-SW-Department (10.0.0.171) - Switch logs

Step 5.2: Test pfSense Firewall Logging

1. Generate firewall blocks:

From any workstation, attempt to access a blocked port:

```
bash  
telnet 8.8.8.8 23
```

2. Search for blocked traffic in Splunk:

```
index=network host="10.0.0.1" action=block  
| table _time, src_ip, dest_ip, dest_port, action  
| head 10
```

Expected results:

- Entries showing blocked outbound traffic
- Source IP = your workstation
- Destination = external IP
- Action = block

3. Generate firewall allows:

Browse to a website:

```
bash  
curl -I https://www.google.com
```

Search for allowed traffic:

```
index=network host="10.0.0.1" action=pass dest_port=443  
| table _time, src_ip, dest_ip, dest_port, action  
| head 10
```

Step 5.3: Test Switch Configuration Changes Logging

1. Make a configuration change on a switch:

```
cisco
```

```
configure terminal
interface Ethernet0/0
description Test-Configuration-Change-Logging
exit
exit
write memory
```

2. Search for the event in Splunk:

```
index=network host="10.0.0.161" "Configuration"
| head 5
```

Expected result:

```
%SYS-5-CONFIG_I: Configured from console by admin on vty0
```

Step 5.4: Test SSH Access Logging

1. SSH to a switch from Jump Server:

```
bash

ssh admin@10.0.0.161
```

2. Exit immediately:

```
exit
```

3. Search for SSH login in Splunk:

```
index=network (host="10.0.0.161") (SSH OR "LOGIN")
| table _time, host, _raw
```

Expected results:

- SSH session opened
- User login successful
- SSH session closed

Step 5.5: Test ACL Deny Logging

1. Attempt SSH from unauthorized IP (not Jump Server):

From IT workstation (10.26.0.x):

```
bash
ssh admin@10.0.0.161
```

2. Search for denied attempts in Splunk:

```
index=network host="10.0.0.161" "denied" OR "IPACCESSLOGP"
| table _time, src_ip, dest_ip, _raw
```

Expected result:

```
%SEC-6-IPACCESSLOGP: list 10 denied tcp 10.26.0.100(54321) -> 10.0.0.161(22), 1 packet
```

This confirms:

- ☒ ACL is blocking unauthorized access
- ☒ Denied attempts are being logged
- ☒ Security monitoring is working

Step 5.6: Performance Verification

1. Check Splunk indexing performance:

```
index=network
| stats count by _indextime span=1h
| timechart span=1h sum(count) as events_per_hour
```

Expected results:

- Steady event flow
- No gaps in data
- Performance within acceptable limits

2. Check index disk usage:

Navigate to: **Settings** → **Indexes**

Verify:

- **network** index: Size is growing
- **Retention:** 90 days

- **Size limit:** 5000 MB not exceeded

3. Check Splunk system health:

Navigate to: **Settings** → **Server Settings** → **Health Report**

Verify:

- **Splunk status:** Green
 - **Indexing rate:** Acceptable
 - **Search performance:** Normal
-

Security Best Practices

Access Control

1. Jump Server Architecture (Zero Trust)

Implemented:

- All administrative access through Jump Server (10.0.0.50)
- No direct access to infrastructure from user VLANs
- Centralized access control and audit logging

Benefits:

- Single point of authentication
- Comprehensive audit trail
- Simplified security monitoring
- Reduced attack surface

2. SSH Hardening

Implemented on all switches:

- SSH version 2 only (v1 disabled)
- Strong passwords (15+ characters)
- Access Control Lists (ACL 10) restricting source IPs
- Session timeouts (10 minutes)
- Limited authentication retries (2 attempts)

- 2048-bit RSA keys

3. Least Privilege Access

✓ Implemented:

- Separate admin accounts per device
- Privilege level 15 only for authorized administrators
- Local authentication (no shared credentials)

Network Security

1. VLAN Segmentation

✓ Implemented:

- Management VLAN (10) isolated from user VLANs
- Security monitoring infrastructure on dedicated VLAN
- Inter-VLAN routing controlled by pfSense

2. Firewall Rules

✓ Implemented:

- Default deny between VLANs
- Explicit allow rules for necessary traffic
- IT VLAN → Splunk web interface (port 8000)
- All VLANs → Syslog denied (only devices send logs)

3. Encrypted Communications

✓ Implemented:

- SSH for switch management (encrypted CLI access)
- HTTPS for Splunk web interface (optional: configure SSL cert)
- Syslog over UDP (Note: plaintext, consider syslog-ng with TLS for production)

Logging Security

1. Centralized Log Storage

✓ Implemented:

- All network devices → Splunk SIEM

- 90-day retention
- Tamper-resistant (logs off-device)

2. Log Integrity

Recommendations for production:

- Enable log forwarding redundancy (multiple syslog servers)
- Implement log integrity checking (SIEM correlation rules)
- Regular backup of Splunk indexes
- Consider WORM storage for compliance

3. Sensitive Data

Implemented:

- Logs stored in Management VLAN (restricted access)
- Splunk access controlled via Jump Server + pfSense rules
- Role-Based Access Control (RBAC) in Splunk (configure as needed)

Operational Security

1. Configuration Management

✅ Implemented:

- write memory after every switch change
- Configuration change logging enabled
- Splunk alerting on configuration changes (can be configured)

Recommendations:

- Maintain configuration backups (TFTP/SFTP)
- Version control for switch configs
- Change management process for production

2. Patch Management

Recommendations:

- Keep Splunk updated (currently 10.0.2)
- Apply pfSense security updates monthly
- Cisco IOS updates for security vulnerabilities

- Ubuntu security patches (unattended-upgrades)

3. Monitoring & Alerting

Implemented:

- Real-time log ingestion
- Searchable 90-day history
- All security events logged

Recommendations (next steps):

- Create Splunk alerts for:
 - Failed SSH attempts (threshold: 5 in 10 minutes)
 - Configuration changes on critical devices
 - Firewall rule modifications
 - Unusual traffic patterns
 - Set up email notifications for critical alerts
 - Weekly security review of Splunk searches
-

Troubleshooting

Splunk Server Issues

Problem: Splunk web interface not accessible

Symptoms:

- Browser shows "Connection refused" or timeout
- Cannot access <http://10.0.0.75:8000>

Diagnosis:

```
bash
```

Check if Splunk is running

```
sudo /opt/splunk/bin/splunk status
```

Check if port 8000 is listening

```
sudo ss -tlnp | grep 8000
```

Check firewall rules

```
sudo ufw status
```

Solutions:

1. Splunk not running:

bash

```
sudo /opt/splunk/bin/splunk start
```

2. Listening on localhost only (127.0.0.1):

bash

```
sudo /opt/splunk/bin/splunk set web-port 8000 -httpport 8000:0.0.0.0
```

```
sudo /opt/splunk/bin/splunk restart splunkweb
```

3. Firewall blocking (Ubuntu UFW):

bash

```
sudo ufw allow 8000/tcp
```

4. pfSense firewall rule missing:

- Verify rule exists: Firewall → Rules → VLAN26_IT_VLAN
- Add rule allowing TCP/8000 from IT VLAN to 10.0.0.75

Problem: No logs appearing in Splunk

Symptoms:

- Search `index=network` returns 0 results
- Devices configured but no data received

Diagnosis:

bash


```
# Check if UDP 514 is listening
```

```
sudo ss -ulnp | grep 514
```

```
# Check Splunk input configuration
```

```
sudo /opt/splunk/bin/splunk list inputstatus
```

```
# Check for incoming packets
```

```
sudo tcpdump -i ens3 udp port 514 -n
```

Solutions:

1. UDP input not configured:

- Settings → Data inputs → UDP → Add new
- Port: 514, Sourcetype: syslog, Index: network

2. Firewall blocking syslog:

```
bash
```

```
sudo ufw allow 514/udp
```

3. Source devices not configured:

- Verify pfSense: Status → System Logs → Settings → Remote Logging
- Verify switches: `show logging` (should show 10.0.0.75)

4. Network connectivity issue:

```
bash
```

```
# From Splunk server, test connectivity to source
```

```
ping 10.0.0.1 # pfSense
```

```
ping 10.0.0.161 # Switch
```

pfSense Integration Issues

Problem: pfSense not sending logs

Diagnosis on pfSense:

1. Navigate to **Status** → **System Logs** → **Settings**
2. Verify "Remote Logging Options":
 - ✓ Send log messages to remote syslog server
 - Remote log servers: `10.0.0.75:514`

3. Check system log: **Status** → **System Logs** → **System**

- Look for: "Syslog started: remote syslog server(s) added"

Solutions:

1. Configuration not saved:

- Re-enter settings and click Save

2. Network connectivity:

```
bash

# From pfSense shell/console
ping 10.0.0.75
```

3. Test syslog manually:

```
bash

# From pfSense shell
logger -t test "Test syslog message to Splunk"
```

Search in Splunk: `index=network "Test syslog"`

Problem: Firewall events missing

Diagnosis:

```
index=network host="10.0.0.1"
| stats count by sourcetype
```

If only seeing system events, not filterlog:

Solution:

1. Verify "Firewall Events" is checked in Remote Syslog Contents
2. Generate test traffic (ping, web browsing)
3. Check pfSense firewall log locally: **Status** → **System Logs** → **Firewall**
4. If local logs exist but not in Splunk:
 - Wait 5 minutes for buffering
 - Restart syslog: `pfSsh.php playback svc restart syslogd`

Cisco Switch Issues

Problem: Cannot SSH to switches

Symptoms:

- `ssh: connect to host 10.0.0.161 port 22: Connection refused`
- OR: `no matching key exchange method found`

Solution 1: Connection refused (SSH not enabled)

Console into switch:

```
cisco

show ip ssh

! If SSH not enabled:
crypto key generate rsa modulus 2048
ip ssh version 2
```

Solution 2: Key exchange method error

From Jump Server:

```
bash

ssh -oKexAlgorithms=+diffie-hellman-group14-sha1 -oHostKeyAlgorithms=+ssh-rsa admin@10.0.0.161
```

Or update `/etc/ssh/ssh_config` (see Step 4.1)

Solution 3: ACL blocking even from Jump Server

```
cisco

show access-lists 10

! Verify 10.0.0.50 is permitted
! If wrong IP:
configure terminal
no access-list 10
access-list 10 permit host 10.0.0.50
access-list 10 deny any log
```

Problem: Switch not sending syslog to Splunk

Diagnosis on switch:

```
cisco
```

```
show logging
```

```
! Look for:
```

```
! Logging to 10.0.0.75 (enabled)
```

```
! Trap logging: level informational
```

Solutions:

1. Logging not configured:

```
cisco
```

```
configure terminal
```

```
logging host 10.0.0.75
```

```
logging trap informational
```

```
logging on
```

2. Source interface not set:

```
cisco
```

```
logging source-interface Vlan10
```

3. Test connectivity:

```
cisco
```

```
ping 10.0.0.75
```

4. Generate test log:

```
cisco
```

```
configure terminal
```

```
interface Ethernet0/0
```

```
description Test-Syslog-Message
```

```
exit
```

```
exit
```

Search Splunk: `(index=network host="10.0.0.161" "Test-Syslog")`

Network Connectivity Issues

Problem: No route to host / Network unreachable

Diagnosis:

```
bash

# From Splunk server
ip route
ping 10.0.0.1 # Gateway

# From switch
show ip route
```

Solutions:

1. Missing default gateway on Splunk:

```
bash

# Check netplan config
cat /etc/netplan/00-installer-config.yaml

# Should have:
routes:
  - to: default
    via: 10.0.0.1
```

2. VLAN misconfiguration on switch:

```
cisco

! Verify switch port is in correct VLAN
show vlan id 10
show interfaces Ethernet2/1 switchport

! Should show: Access Mode VLAN: 10
```

3. pfSense routing issue:

- Verify VLAN 10 interface exists on pfSense
 - Check: Interfaces → VLAN10_MANAGEMENT_VLAN
 - IP should be: 10.0.0.1/24
-

Appendix

Appendix A: Implementation Summary

Deployment Date: December 6, 2025
Implemented By: Prageeth Panicker
Environment: EVE-NG virtualized network lab

Devices Configured:

Device	Role	IP Address	Status
Splunk-SIEM	SIEM Server	10.0.0.75	✔ Operational
pfSense-HA	Firewall	10.0.0.1/2	✔ Logging
Access-SW-Servers	Access Switch	10.0.0.161	✔ Logging
Core-SW1	Core Switch	10.0.0.120	✔ Logging
Core-SW2	Core Switch	10.0.0.121	✔ Logging
Access-SW-Department	Access Switch	10.0.0.171	✔ Logging
Jump-Server	SSH Gateway	10.0.0.50	✔ Access Control

Logs Collected (First 24 Hours):

- **Total Events:** 31,000+
- **pfSense:** 30,977 events (firewall, system, DHCP)
- **Switches:** 15 events (configuration, SSH, system)

Security Controls Implemented:

- ✔ Zero Trust architecture (Jump Server gateway)
- ✔ SSH version 2 only on all devices
- ✔ ACL restricting switch access to Jump Server
- ✔ Centralized logging to SIEM
- ✔ 90-day log retention
- ✔ Role-based access control

Appendix B: Quick Reference Commands

Splunk Server

```
bash
```

Start/Stop/Restart Splunk

```
sudo /opt/splunk/bin/splunk start
```

```
sudo /opt/splunk/bin/splunk stop
```

```
sudo /opt/splunk/bin/splunk restart
```

Check Splunk status

```
sudo /opt/splunk/bin/splunk status
```

View Splunk processes

```
ps aux | grep splunk
```

Check listening ports

```
sudo ss -tlnp | grep splunk
```

View Splunk logs

```
tail -f /opt/splunk/var/log/splunk/splunkd.log
```

Search from CLI (rare, usually use web UI)

```
/opt/splunk/bin/splunk search "index=network" -auth admin:password
```

pfSense

```
bash
```

Restart syslog daemon

```
/usr/local/etc/rc.d/syslogd restart
```

Test syslog manually

```
logger -t test "Test message"
```

View local firewall log

```
clog /var/log/filter.log | tail -20
```

Check syslog configuration

```
grep "loghost" /var/etc/syslog.conf
```

Cisco Switches

```
cisco
```

```
! Check SSH status
show ip ssh
show ssh

! Check logging status
show logging

! Check running config sections
show running-config | include logging
show running-config | section line vty
show access-lists 10

! Test connectivity
ping 10.0.0.75

! Generate test log
configure terminal
interface Loopback0
description Test-Log-Generation
exit
exit

! Save configuration
write memory

! Backup configuration to TFTP (optional)
copy running-config tftp://10.0.0.75/switch-backup.cfg
```

Jump Server

```
bash

# SSH to network devices
ssh admin@10.0.0.161 # Switch
ssh admin@10.0.0.1 # pfSense

# Monitor syslog traffic
sudo tcpdump -i ens3 udp port 514 -n -A

# Test connectivity
ping 10.0.0.75 # Splunk
ping 10.0.0.1 # pfSense gateway
```

Appendix C: Useful Splunk Searches

Network Overview


```
index=network
| stats count by host, sourcetype
| sort - count
```

Firewall Traffic Analysis

```
index=network host="10.0.0.1" sourcetype=syslog
| rex field=_raw "(?<action>block|pass)"
| stats count by action, src_ip, dest_port
| sort - count
```

SSH Login Monitoring

```
index=network (SSH OR "LOGIN" OR "SEC-6")
| table _time, host, user, src_ip, action, _raw
| sort - _time
```

Top Talkers (Most Active IPs)

```
index=network host="10.0.0.1"
| rex field=_raw "(?<src_ip>\d+\.\d+\.\d+\.\d+),(?<dest_ip>\d+\.\d+\.\d+\.\d+)"
| stats count by src_ip
| sort - count
| head 10
```

Configuration Changes

```
index=network ("CONFIG" OR "configuration")
| table _time, host, user, _raw
| sort - _time
```

Failed Authentication Attempts

```
index=network ("failed" OR "denied" OR "invalid")
| stats count by host, src_ip, user
| sort - count
```

Events per Hour (Performance Monitoring)

index=network
| timechart span=1h count by host

Appendix D: Network Diagram



VLAN 30 (Server VLAN - 10.30.0.0/24)

DC01 (10.30.0.10) - Domain Controller

FILE-SRV01 (10.30.0.20) - File Server

APP-SRV01 (10.30.0.30) - Application Server

Appendix E: Security Checklist

Post-Implementation Security Review:

- ☐ Splunk admin password changed from default
- ☐ Splunk web interface accessible only from authorized VLANs
- ☐ All switch admin passwords set to strong passphrases
- ☐ SSH version 2 enforced on all devices
- ☐ ACLs applied to all switch VTY lines
- ☐ Only Jump Server can SSH to infrastructure devices
- ☐ All configuration changes logged to Splunk
- ☐ 90-day log retention configured
- ☐ Indexes have size limits set
- ☐ pfSense logging all firewall events
- ☐ All 4 switches sending logs to Splunk
- ☐ No Telnet enabled on any device
- ☐ Console access password-protected
- ☐ SNMP community strings changed from defaults (if used)
- ☐ Enable secret configured on all switches
- ☐ Splunk auto-start on boot enabled
- ☐ Regular backups of Splunk indexes planned
- ☐ Configuration backups of all network devices

Appendix F: Future Enhancements

Recommended Next Steps:

1. Wazuh Integration:

- Configure Wazuh → Splunk log forwarding
- Install Wazuh agents on endpoints
- Enable File Integrity Monitoring (FIM)

2. Advanced Alerting:

- Create Splunk alerts for security events

- Email notifications for critical alerts
- Dashboard for SOC operations

3. Compliance:

- CIS Benchmark compliance scanning with Nessus
- GDPR/PCI-DSS log retention policies
- Audit report generation

4. High Availability:

- Splunk indexer clustering (for redundancy)
- Search head clustering
- Backup syslog server

5. Advanced Analytics:

- MITRE ATT&CK framework mapping
- Threat hunting queries
- Anomaly detection with Splunk ML

6. SSL/TLS:

- Enable HTTPS for Splunk web interface
- SSL certificates for production
- Encrypted syslog (syslog-ng with TLS)

Document Revision History

Version	Date	Author	Changes
1.0	Dec 6, 2025	Prageeth Panicker	Initial release - Splunk deployment with pfSense and Cisco switch integration

Acknowledgments

This implementation guide was developed as part of a comprehensive cybersecurity capstone project focusing on enterprise network security monitoring for a mid-sized professional services firm.

Framework References:

- NIST Cybersecurity Framework 2.0
- CIS Controls v8
- MITRE ATT&CK Framework

Tools Used:

- Splunk Enterprise 10.0.2
 - pfSense 2.7+ (Community Edition)
 - Cisco IOS 15.2
 - Ubuntu Server 24.04 LTS
 - EVE-NG (Network Emulation)
-